

COLLOQUY / THOMAS OR TEENSY

One mode shunt – five unchanged body interfaces – Teensy USB powered only

controller

02 CONTROLLER

File: controller.kicad_sch

power

03 POWER

File: power.kicad_sch

harness

04 HARNESS

File: harness.kicad_sch

harness_2

05 HARNESS

File: harness_2.kicad_sch

harness_3

06 HARNESS

File: harness_3.kicad_sch

audio3

07 AUDIO3

File: audio3.kicad_sch

audio3_2

08 AUDIO3

File: audio3_2.kicad_sch

audio4

09 AUDIO4

File: audio4.kicad_sch

audio4_2

10 AUDIO4

File: audio4_2.kicad_sch

audio5

11 AUDIO5

File: audio5.kicad_sch

audio5_2

12 AUDIO5

File: audio5_2.kicad_sch

audio1

13 AUDIO1

File: audio1.kicad_sch

audio1_2

14 AUDIO1

File: audio1_2.kicad_sch

audio2

15 AUDIO2

File: audio2.kicad_sch

audio2_2

16 AUDIO2

File: audio2_2.kicad_sch

io

17 IO

File: io.kicad_sch

teensy

18 TEENSY

File: teensy.kicad_sch

teensy_2

19 TEENSY

File: teensy_2.kicad_sch

dac_power

20 DAC POWER

File: dac_power.kicad_sch

dac_a

21 DAC A

File: dac_a.kicad_sch

dac_a_2

22 DAC A

File: dac_a_2.kicad_sch

dac_b

23 DAC B

File: dac_b.kicad_sch

dac_b_2

24 DAC B

File: dac_b_2.kicad_sch

dac_c

25 DAC C

File: dac_c.kicad_sch

dac_c_2

26 DAC C

File: dac_c_2.kicad_sch

bench

27 BENCH

File: bench.kicad_sch

ears1

28 EARS1

File: ears1.kicad_sch

ears1_2

29 EARS1

File: ears1_2.kicad_sch

ears2

30 EARS2

File: ears2.kicad_sch

ears2_2

31 EARS2

File: ears2_2.kicad_sch

relays

32 RELAYS

File: relays.kicad_sch

relays_2

33 RELAYS

File: relays_2.kicad_sch

relays_3

34 RELAYS

File: relays_3.kicad_sch

logic

35 LOGIC

File: logic.kicad_sch

logic_2

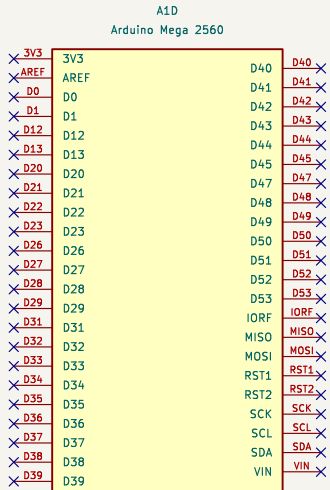
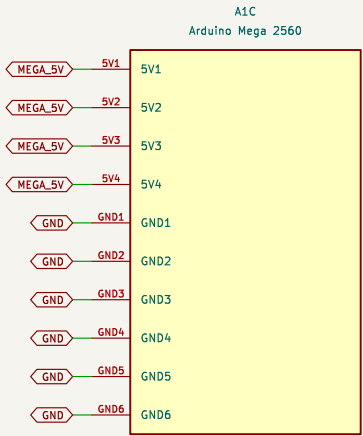
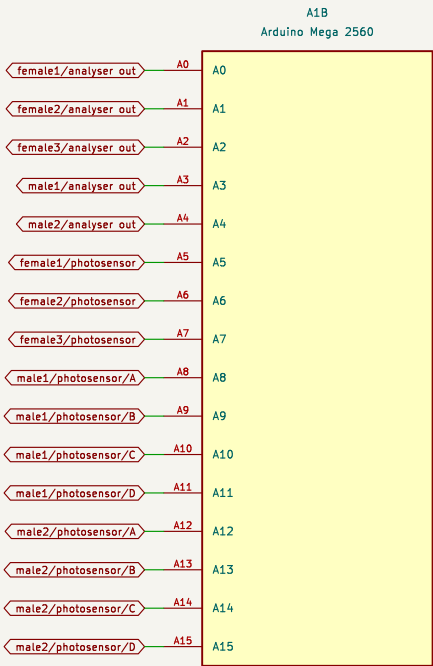
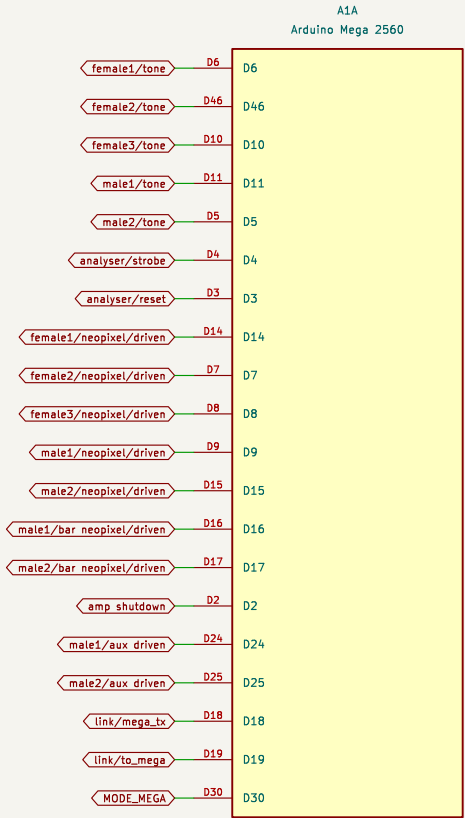
36 LOGIC

File: logic_2.kicad_sch

REVIEW PROTOTYPE: read REVIEW.md and validation reports before fabrication.
NC relay contacts select Thomas. No shunt = Thomas. Teensy 3.3V GPIO are not 5V tolerant.
Protected translators replace LV1T34; powered-off microphone isolation added after first 10K.
Five SMD MSGEQ7N; default attenuation bypasses OPEN. RP1,RP11 remain provisional 10K.

THOMAS / TEENSY – CONTROLLER

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

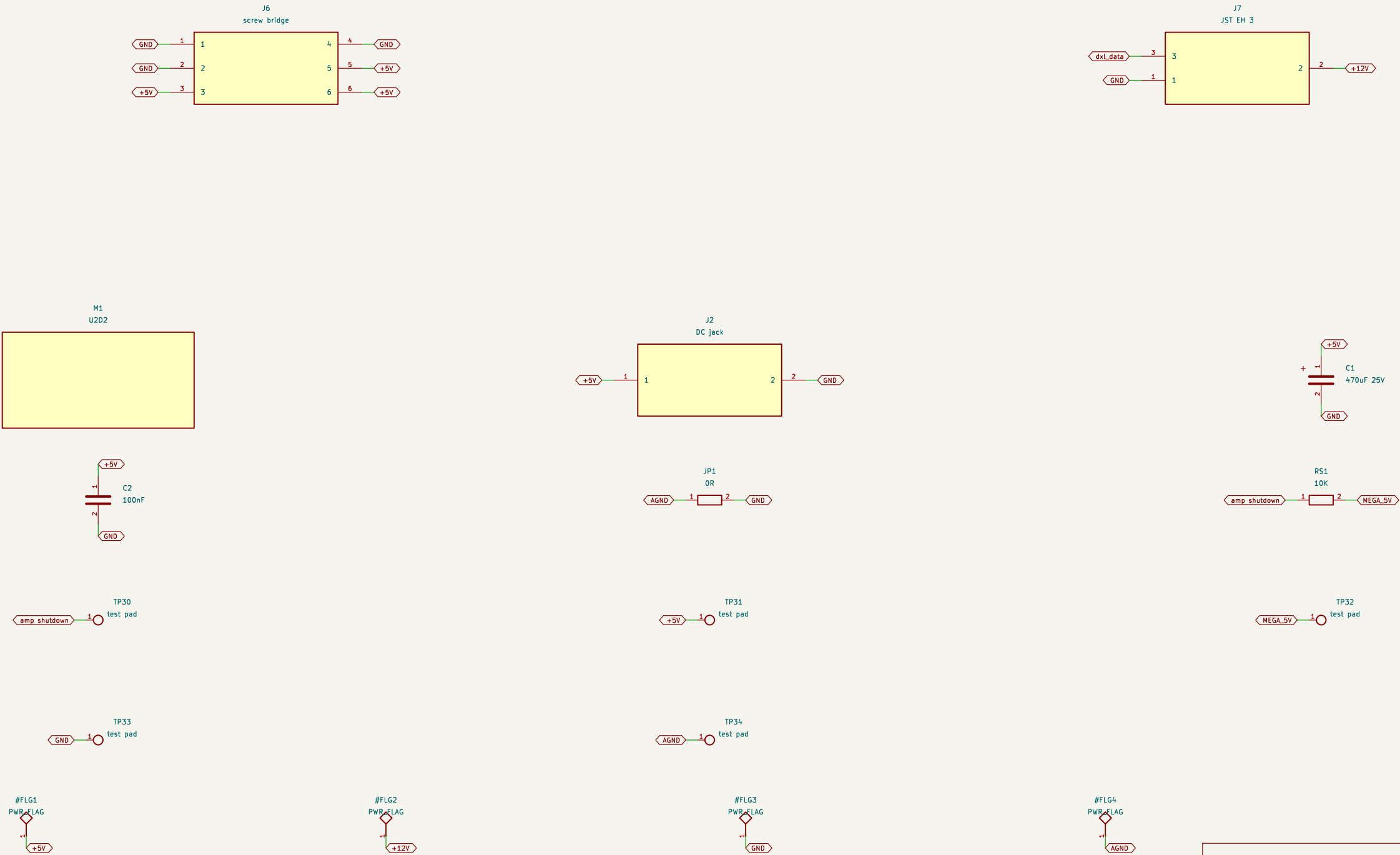
Colloquy of Mobiles

Sheet: /controller/
File: controller.kicad_sch

Title: **CONTROLLER**

Size: A3	Date: 2026-10-01	Rev: A-review
KiCad E.D.A. 9.0.7		Id: 2/36

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /power/
File: power.kicad_sch

Title: POWER

Size: A3 Date: 2026-10-01

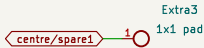
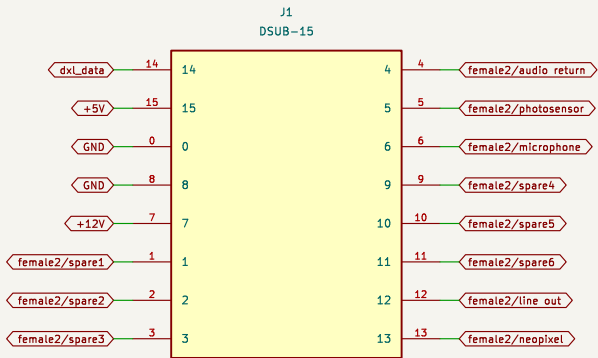
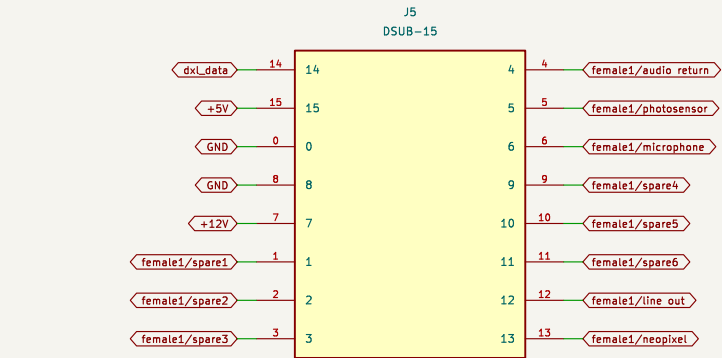
KiCad E.D.A. 9.0.7

Rev: A-review

Id: 3/36

THOMAS / TEENSY – HARNESS

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.

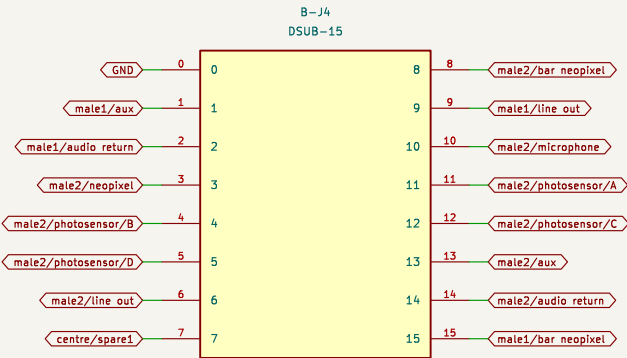
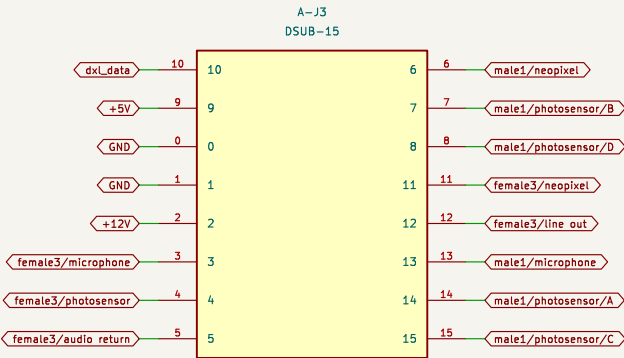


Prototype: verify harness polarity and sensor values before connection
Colloquy of Mobiles
Sheet: /harness/
File: harness.kicad_sch

Title: HARNESS		
Size: A3	Date: 2026-10-01	Rev: A-review
KiCad E.D.A. 9.0.7		Id: 4/36

THOMAS / TEENSY – HARNESS

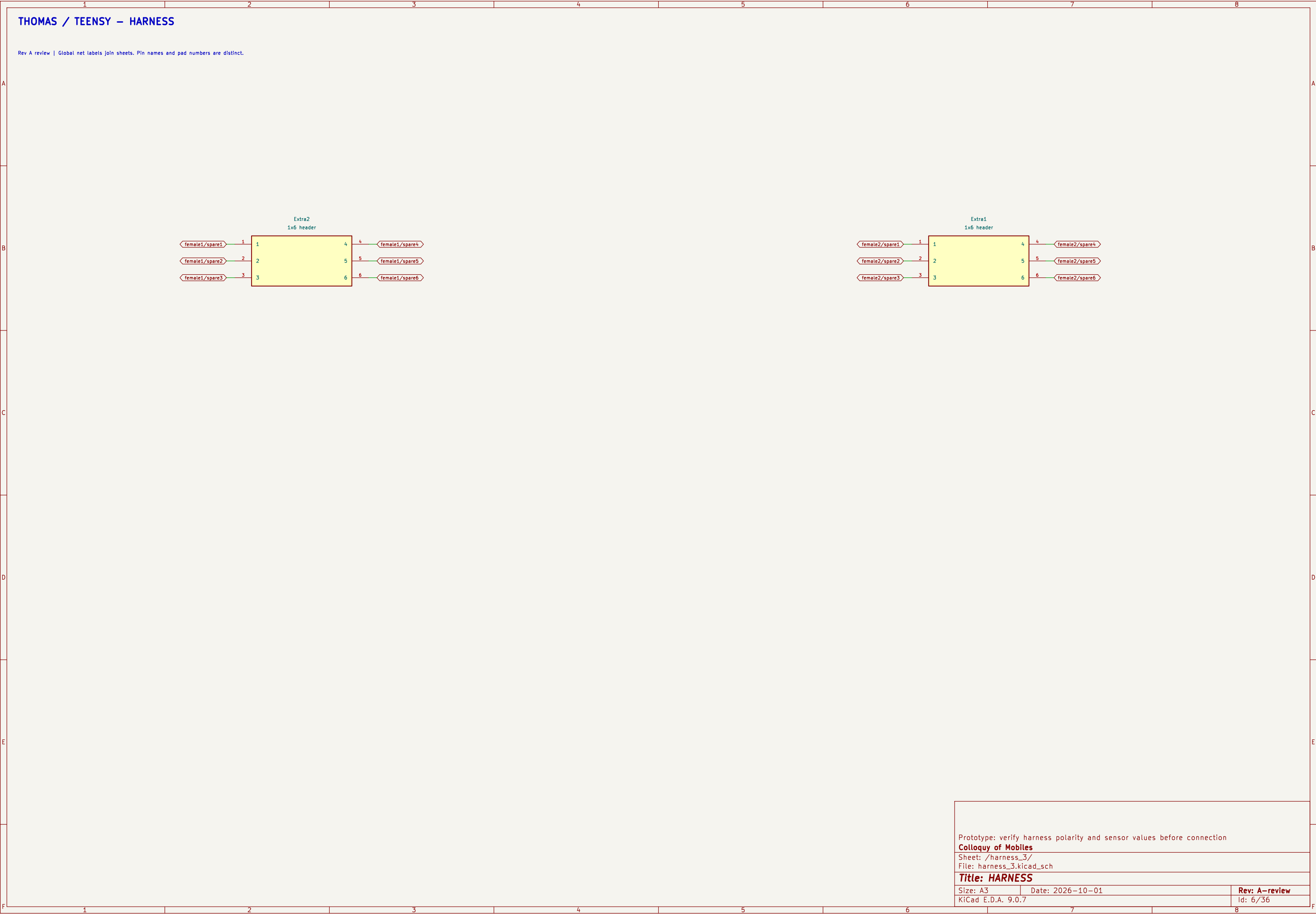
Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection
Colloquy of Mobiles
Sheet: /harness_2/
File: harness_2.kicad_sch

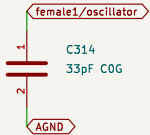
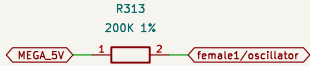
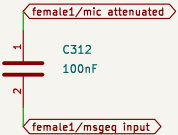
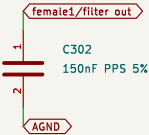
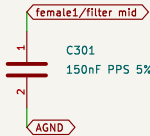
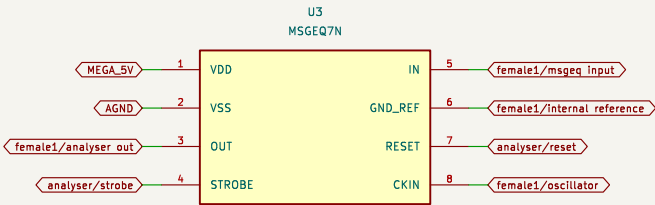
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Size: A3	Date: 2026–10–01	Rev: A–review
KiCad E.D.A. 9.0.7		Id: 5/36



THOMAS / TEENSY – AUDIO3

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /audio3/

File: audio3.kicad_sch

Title: AUDIO3

Size: A3

Date: 2026-10-01

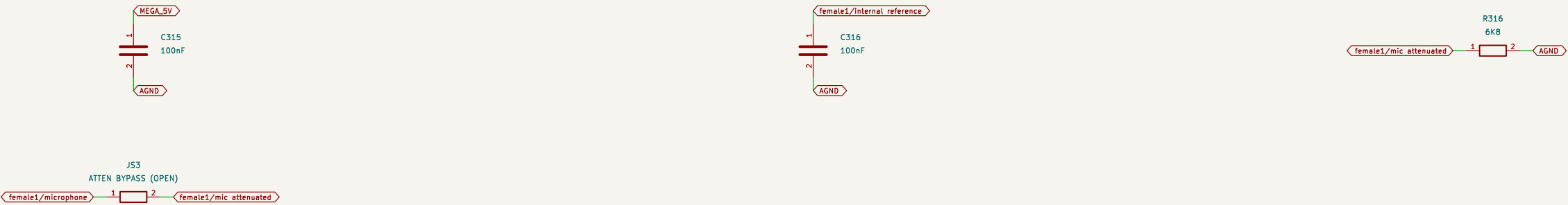
Rev: A-review

KiCad E.D.A. 9.0.7

Id: 7/36

THOMAS / TEENSY – AUDIO3

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.

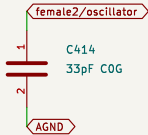
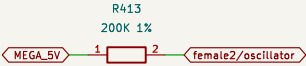
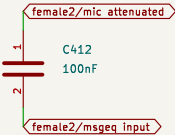
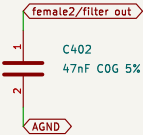
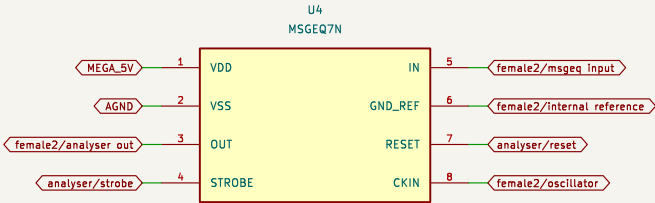


Prototype: verify harness polarity and sensor values before connection
Colloquy of Mobiles
Sheet: /audio3_2/
File: audio3_2.kicad_sch

Title: AUDIO3		
Size: A3	Date: 2026-10-01	Rev: A-review
KiCad E.D.A. 9.0.7		Id: 8/36

THOMAS / TEENSY – AUDIO4

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /audio4/
File: audio4.kicad_sch

Title: AUDIO4

Size: A3
KiCad E.D.A. 9.0.7

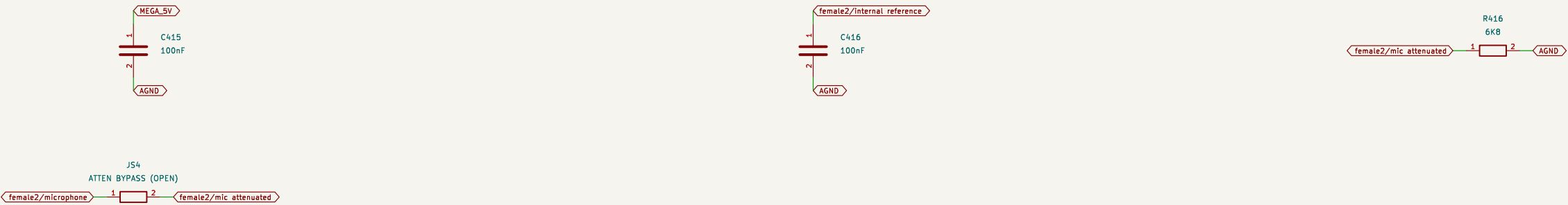
Date: 2026-10-01

Rev: A-review

Id: 9/36

THOMAS / TEENSY – AUDIO4

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

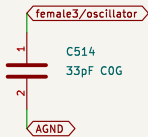
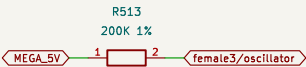
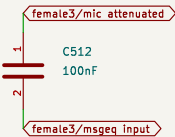
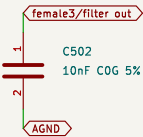
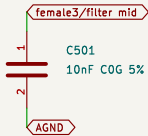
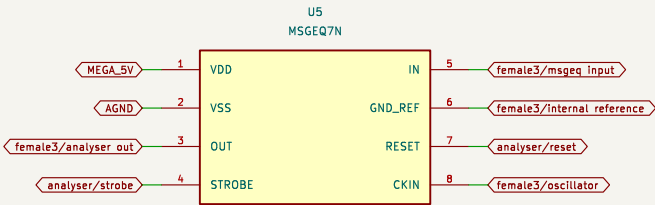
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Title: AUDIO4

Size: A3	Date: 2026-10-01	Rev: A-review
KiCad E.D.A. 9.0.7		Id: 10/36

THOMAS / TEENSY – AUDIO5

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /audio5/

File: audio5.kicad_sch

Title: AUDI05

Size: A3

Date: 2026-10-01

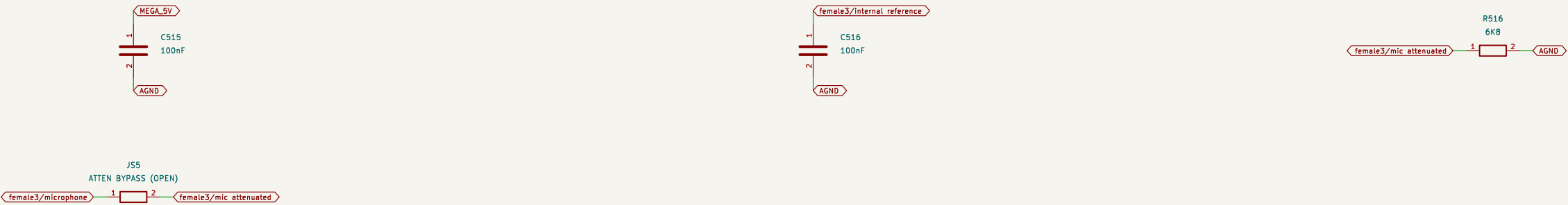
Rev: A-review

KiCad E.D.A. 9.0.7

Id: 11/36

THOMAS / TEENSY – AUDIO5

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.

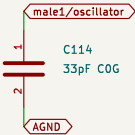
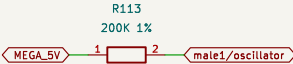
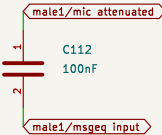
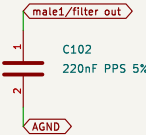
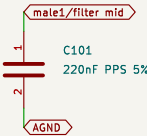
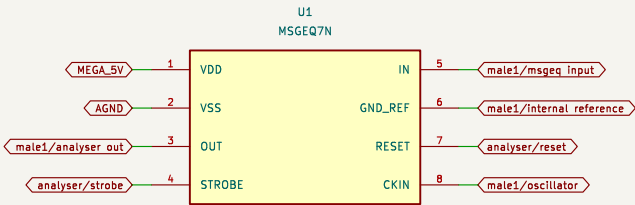


Prototype: verify harness polarity and sensor values before connection
Colloquy of Mobiles
Sheet: /audio5_2/
File: audio5_2.kicad_sch

Title: AUDIO5		
Size: A3	Date: 2026-10-01	Rev: A-review
KiCad E.D.A. 9.0.7		Id: 12/36

THOMAS / TEENSY – AUDIO1

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /audio1/
File: audio1.kicad_sch

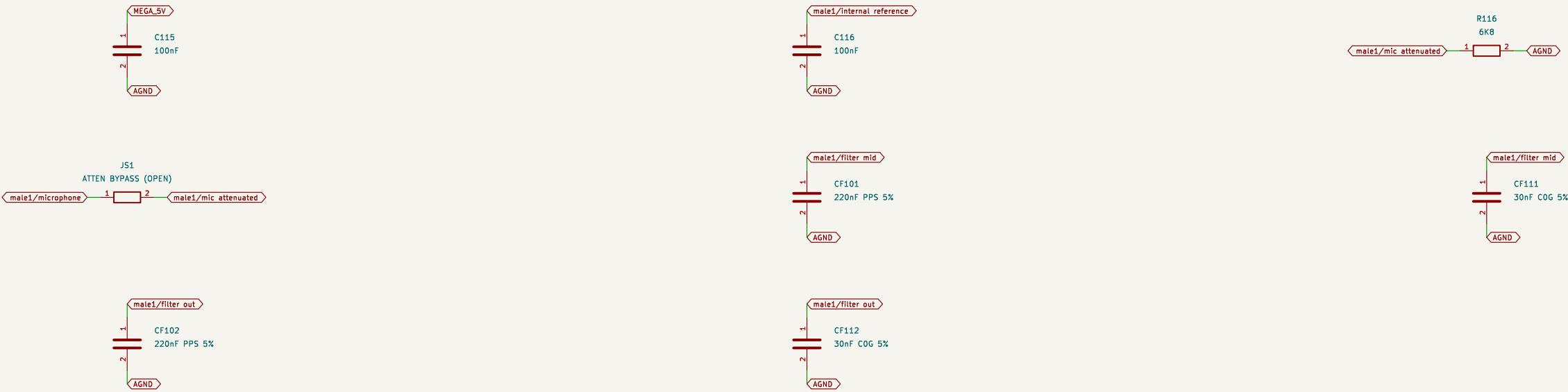
Title: AUDIO1

Size: A3
Date: 2026-10-01

Rev: A-review
Id: 13/36

THOMAS / TEENSY – AUDIO1

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

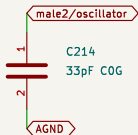
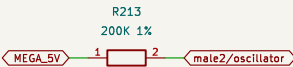
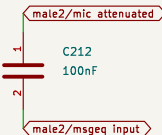
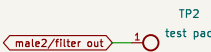
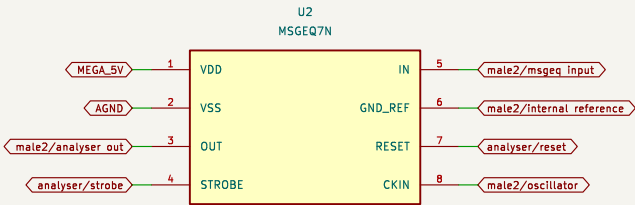
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File: audio1_2.kicad_sch

Title: **AUDIO1**

Size: A3	Date: 2026-10-01	Rev: A-review
KiCad E.D.A. 9.0.7		Id: 14/36

THOMAS / TEENSY – AUDIO2

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /audio2/
File: audio2.kicad_sch

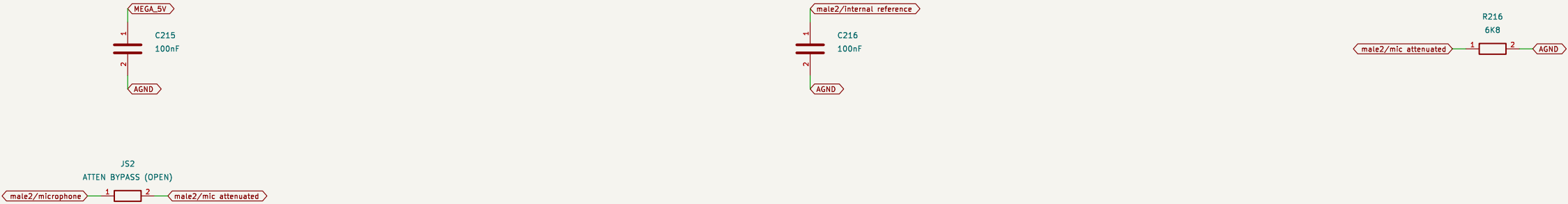
Title: AUDIO2

Size: A3
Date: 2026-10-01

Rev: A-review
Id: 15/36

THOMAS / TEENSY – AUDIO2

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.

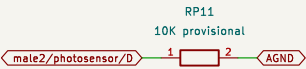
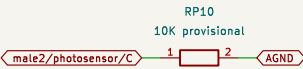
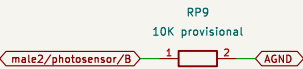
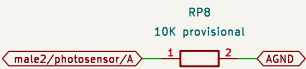
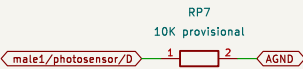
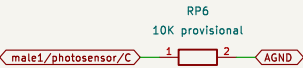
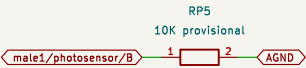
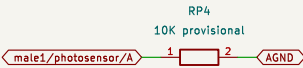


Prototype: verify harness polarity and sensor values before connection
Colloquy of Mobiles
Sheet: /audio2_2/
File: audio2_2.kicad_sch

Title: AUDIO2		
Size: A3	Date: 2026-10-01	Rev: A-review
KiCad E.D.A. 9.0.7		Id: 16/36

THOMAS / TEENSY – IO

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



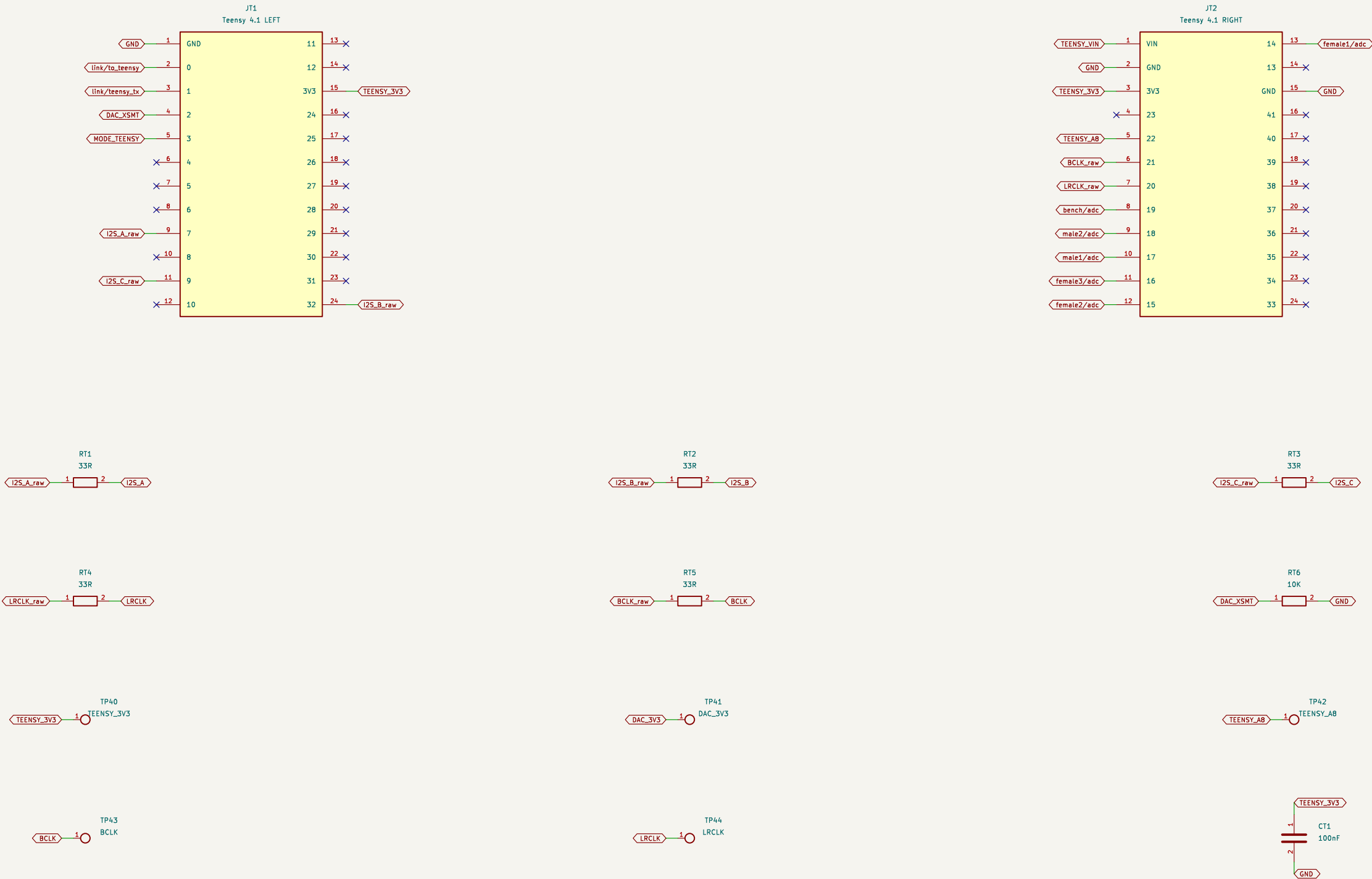
Prototype: verify harness polarity and sensor values before connection
Colloquy of Mobiles
Sheet: /io/
File: io.kicad_sch

Title: IO

Size: A3
Date: 2026-10-01

Rev: A-review
Id: 17/36

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /teensy/
File: teensy.kicad_sch

Title: TEENSY

Size: A3 Date: 2026-10-01

KiCad E.D.A. 9.0.7

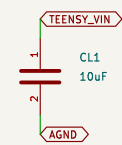
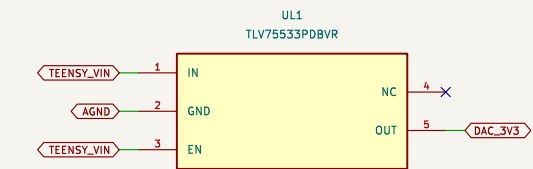
Rev: A-review

Id: 18/36



THOMAS / TEENSY – DAC POWER

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

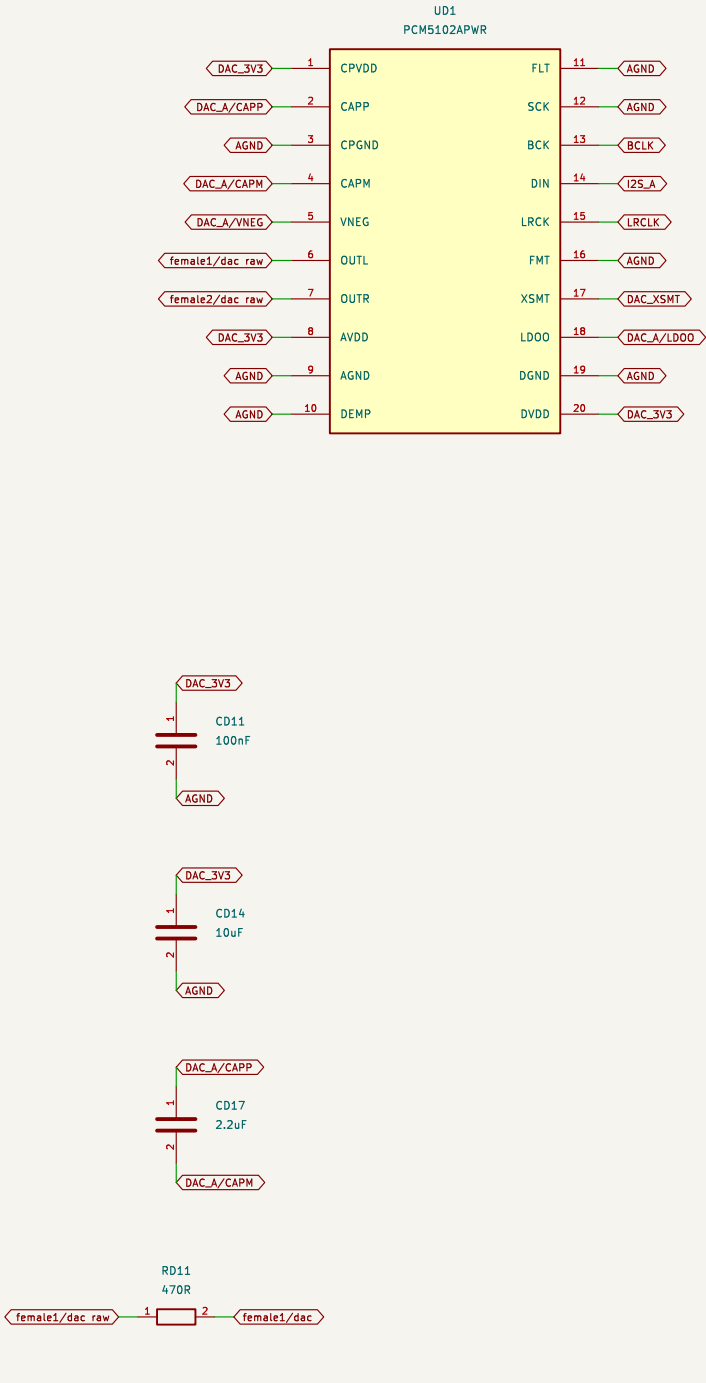
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File: dac_power.kicad_sch

Title: DAC POWER

Size: A3	Date: 2026-10-01	Rev: A-review
KiCad E.D.A. 9.0.7	Id: 20/36	

THOMAS / TEENSY – DAC A

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /dac_a/
File: dac_a.kicad_sch

Title: DAC A

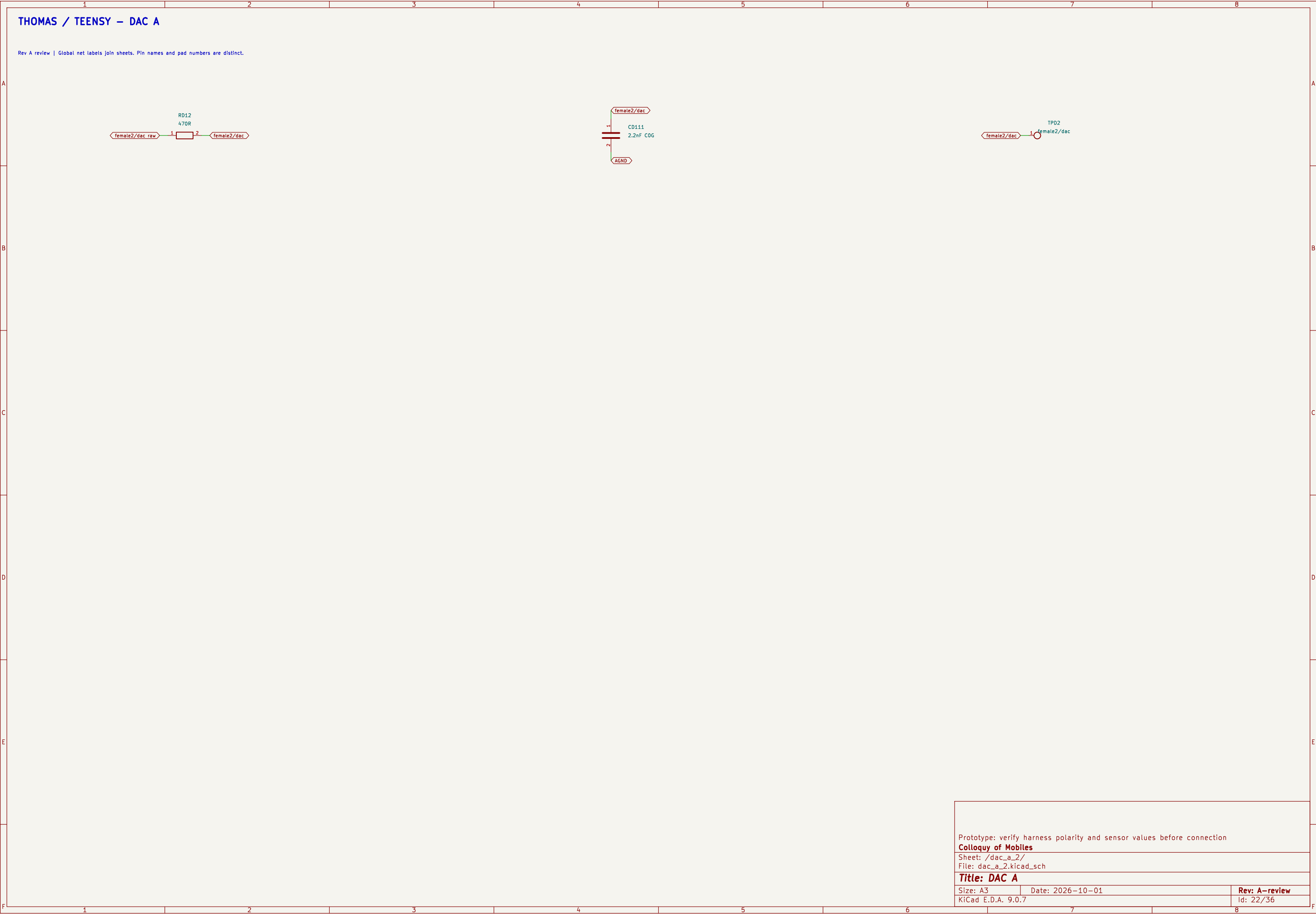
Size: A3

Date: 2026-10-01

Rev: A-review

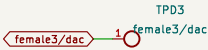
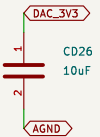
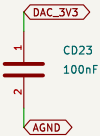
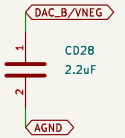
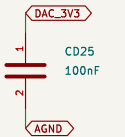
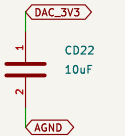
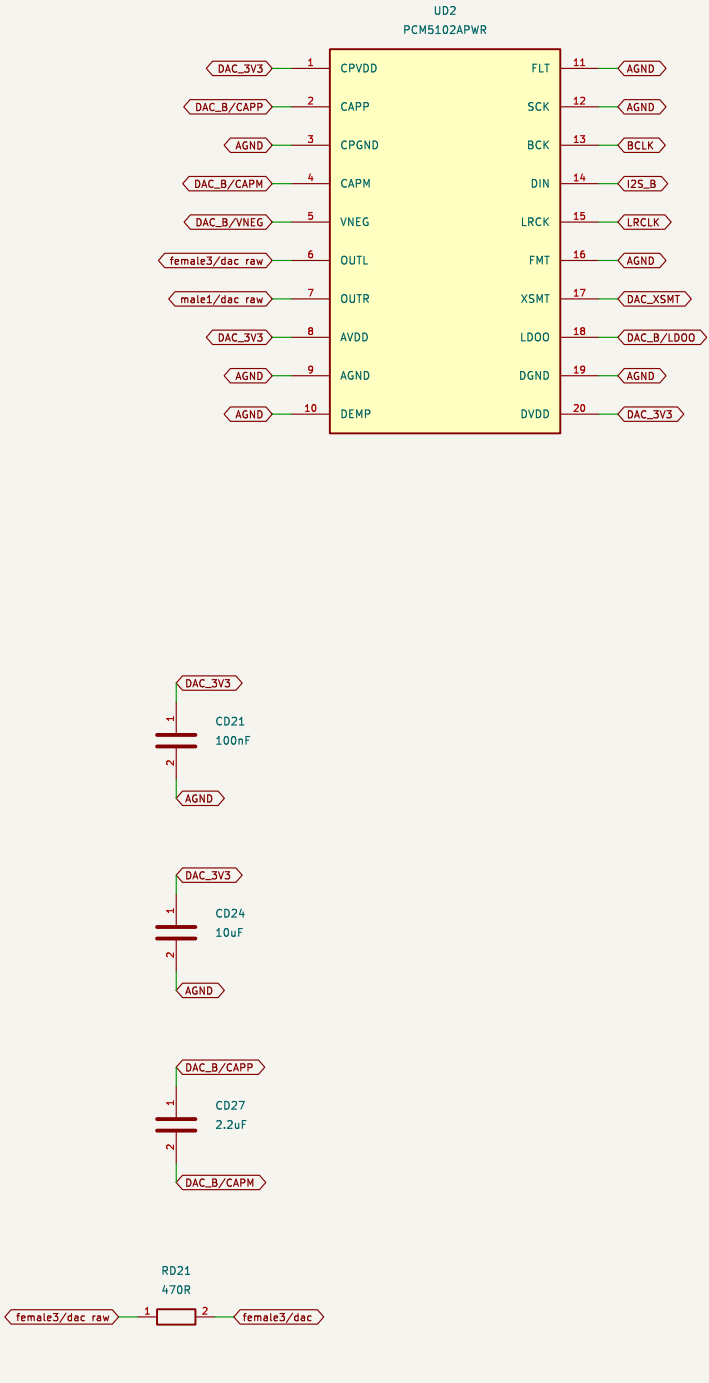
KiCad E.D.A. 9.0.7

Id: 21/36



THOMAS / TEENSY – DAC B

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /dac_b/

File: dac_b.kicad_sch

Title: DAC B

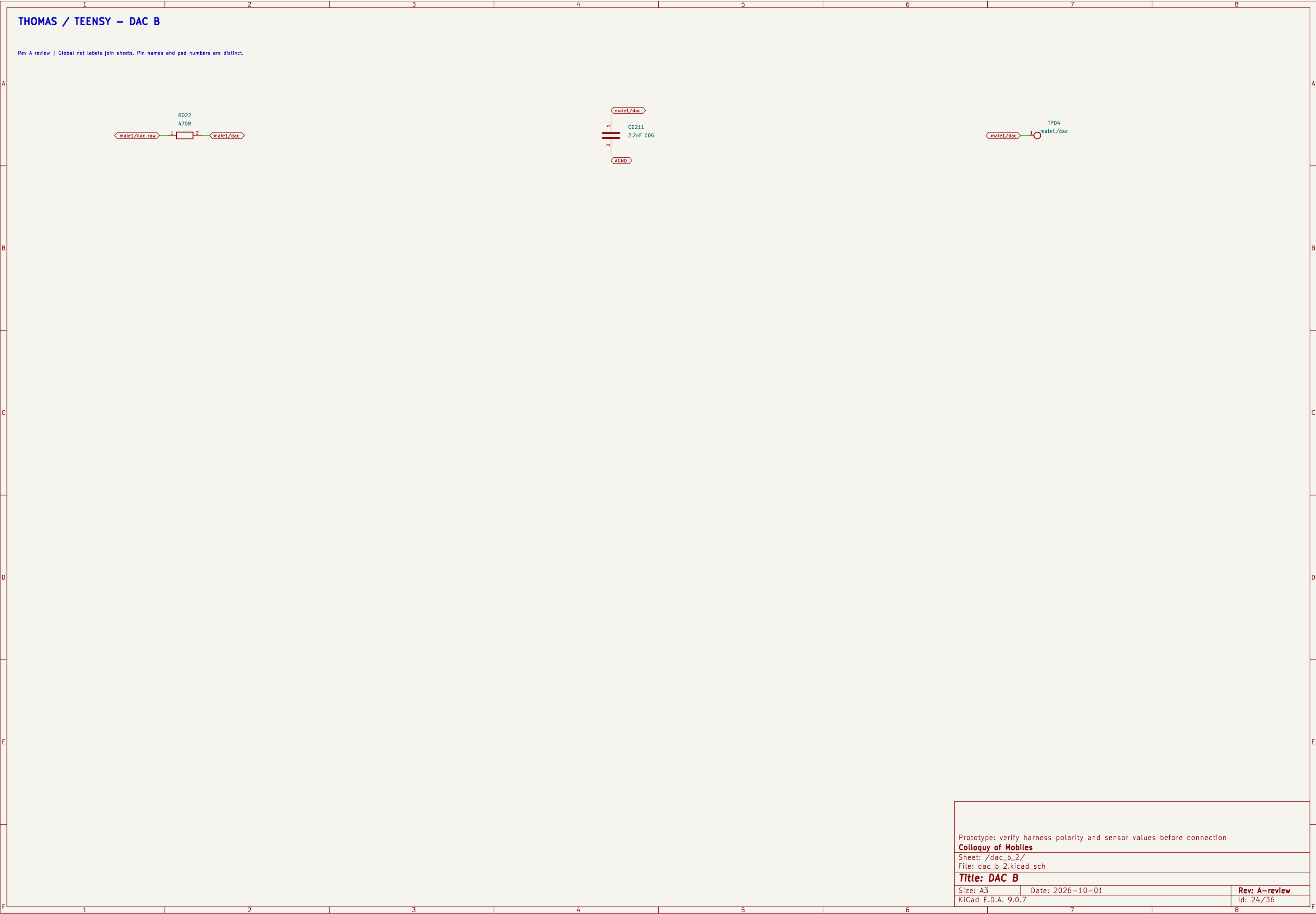
Size: A3

Date: 2026-10-01

Rev: A-review

KiCad E.D.A. 9.0.7

Id: 23/36



A

B

C

D

E

F

A

B

C

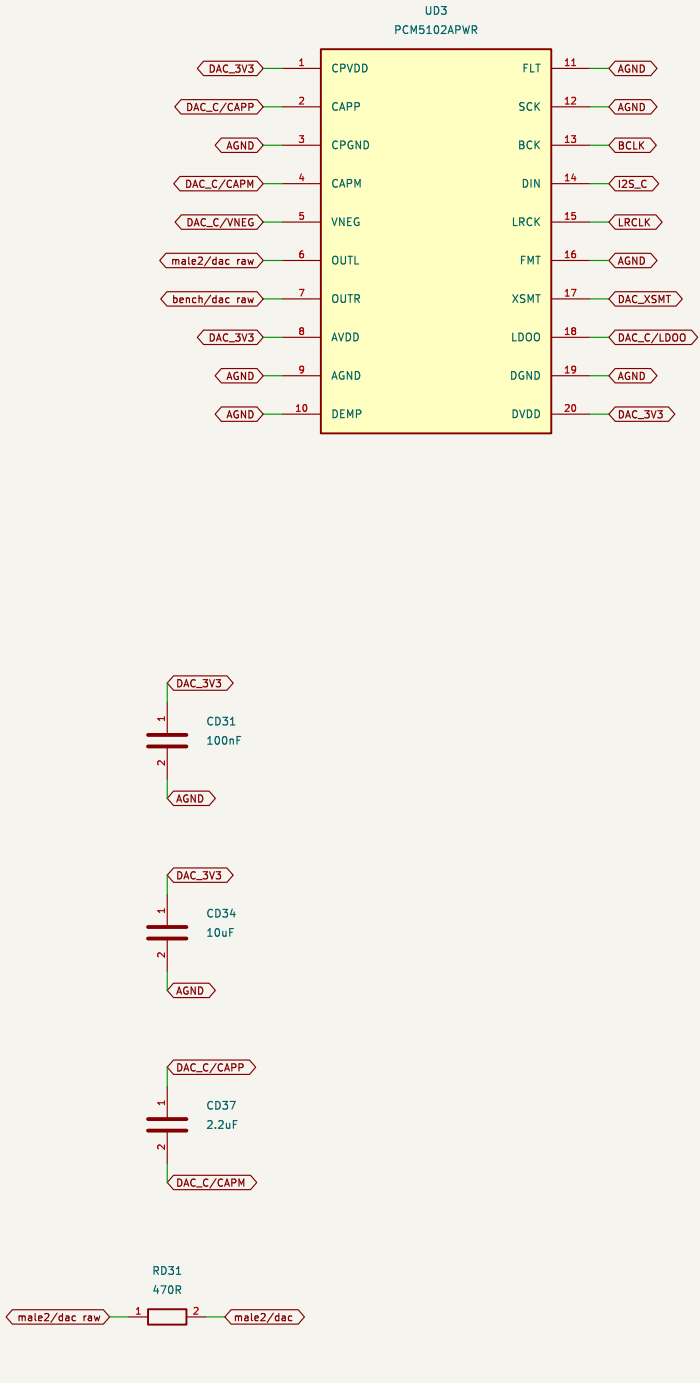
D

E

F

THOMAS / TEENSY – DAC C

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /dac_c/

File: dac_c.kicad_sch

Title: DAC C

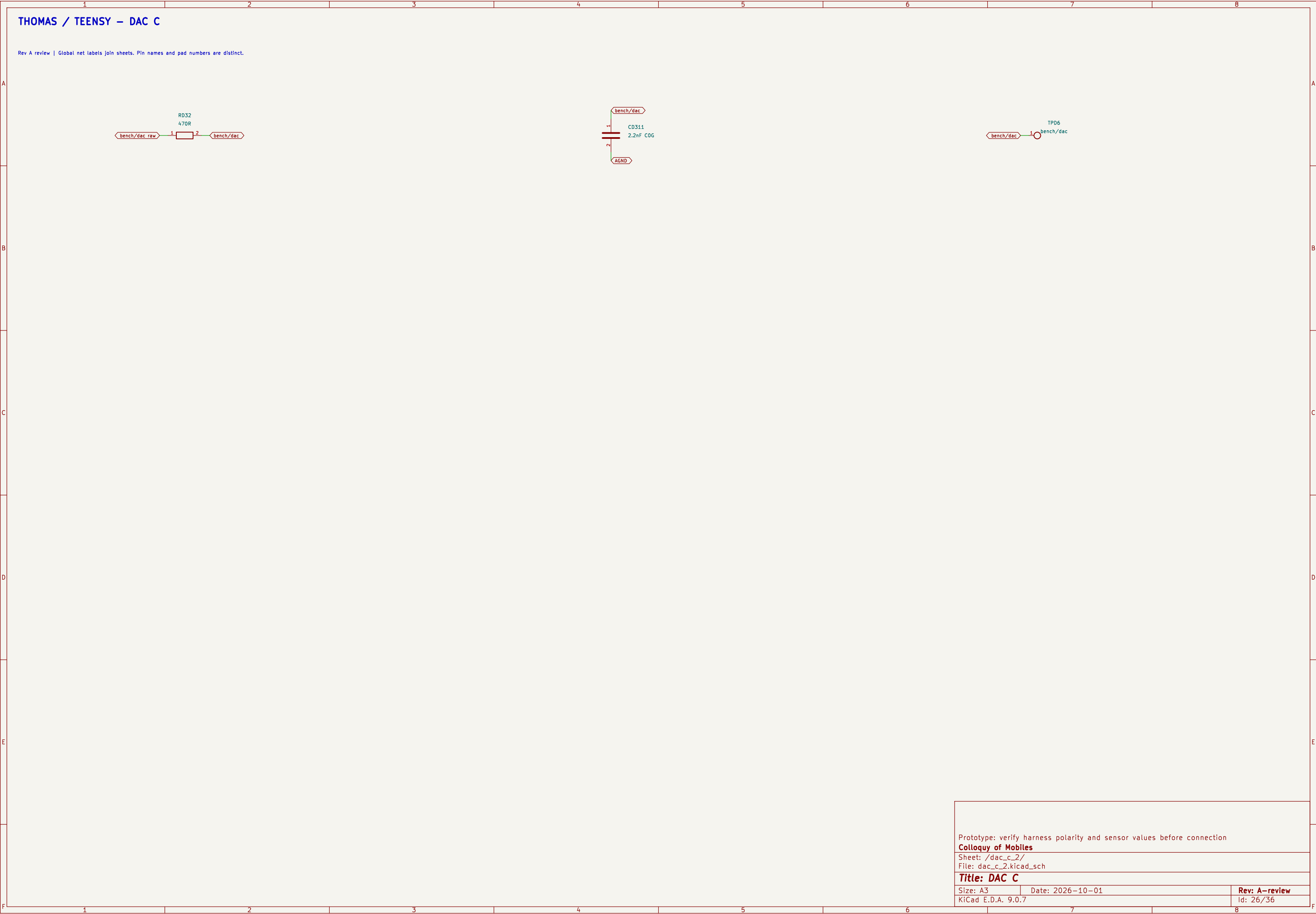
Size: A3

Date: 2026-10-01

Rev: A-review

KiCad E.D.A. 9.0.7

Id: 25/36



A

B

C

D

E

F

A

B

C

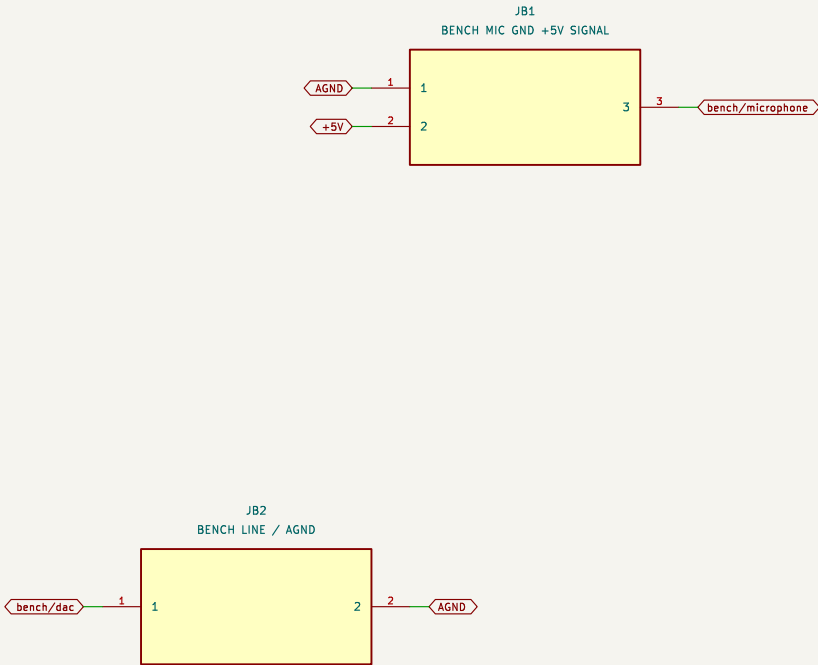
D

E

F

THOMAS / TEENSY – BENCH

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

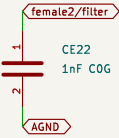
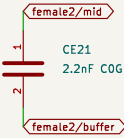
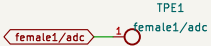
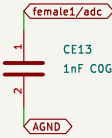
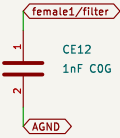
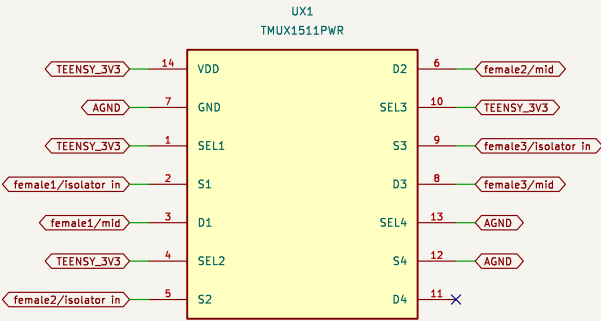
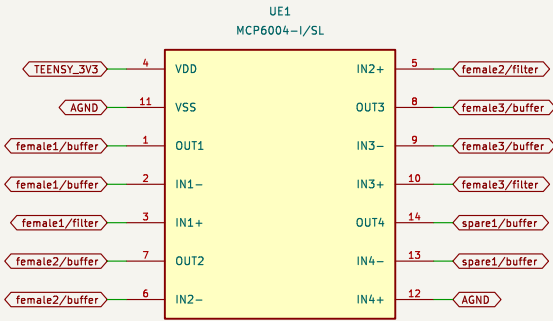
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File: bench.kicad_sch

Title: BENCH

Size: A3	Date: 2026-10-01	Rev: A-review
KiCad E.D.A. 9.0.7	Id: 27/36	

THOMAS / TEENSY – EARS1

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /ears1/

File: ears1.kicad_sch

Title: EARS1

Size: A3

Date: 2026-10-01

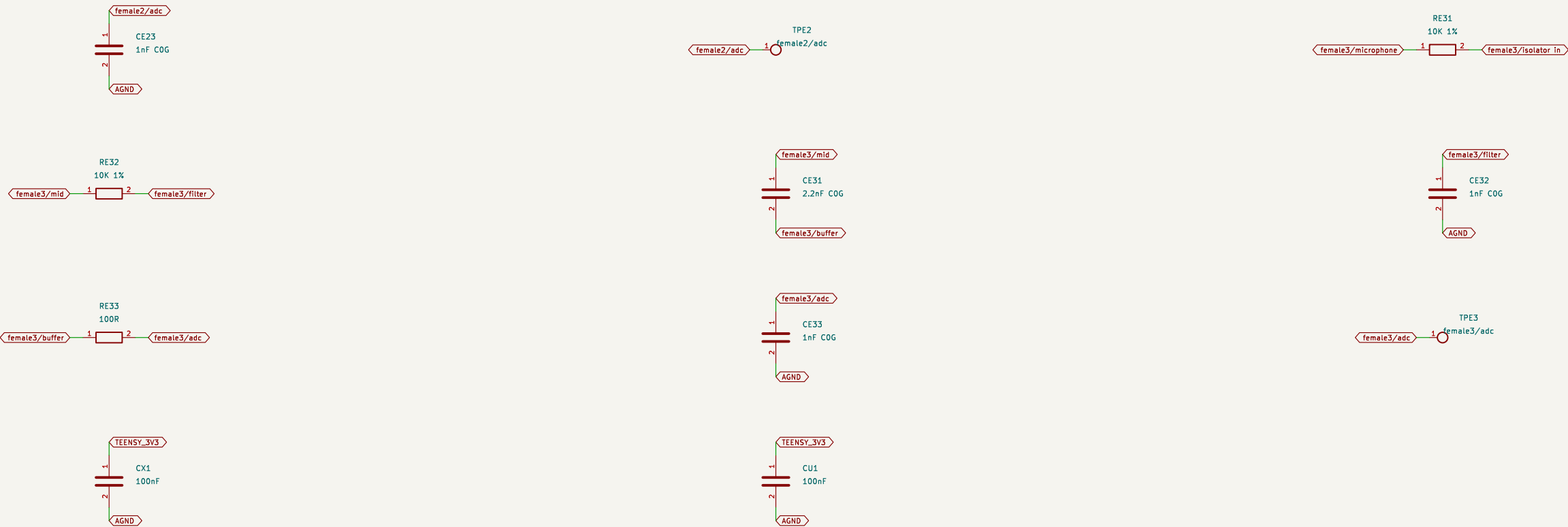
Rev: A-review

KiCad E.D.A. 9.0.7

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THOMAS / TEENSY – EARS1

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.

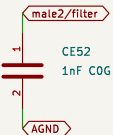
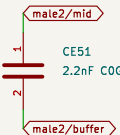
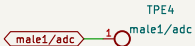
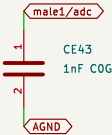
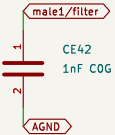
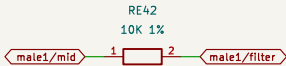
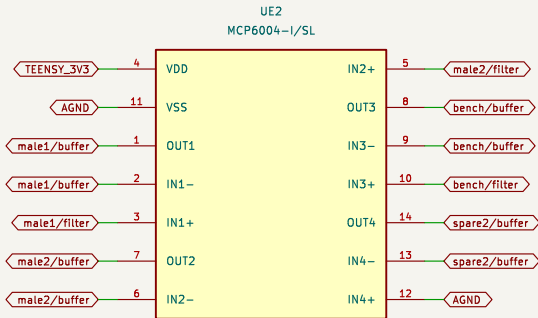


Prototype: verify harness polarity and sensor values before connection
Colloquy of Mobiles
Sheet: /ears1_2/
File: ears1_2.kicad_sch

Title: EARS1		
Size: A3	Date: 2026-10-01	Rev: A-review
KiCad E.D.A. 9.0.7		Id: 29/36

THOMAS / TEENSY – EARS2

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /ears2/

File: ears2.kicad_sch

Title: EARS2

Size: A3

Date: 2026-10-01

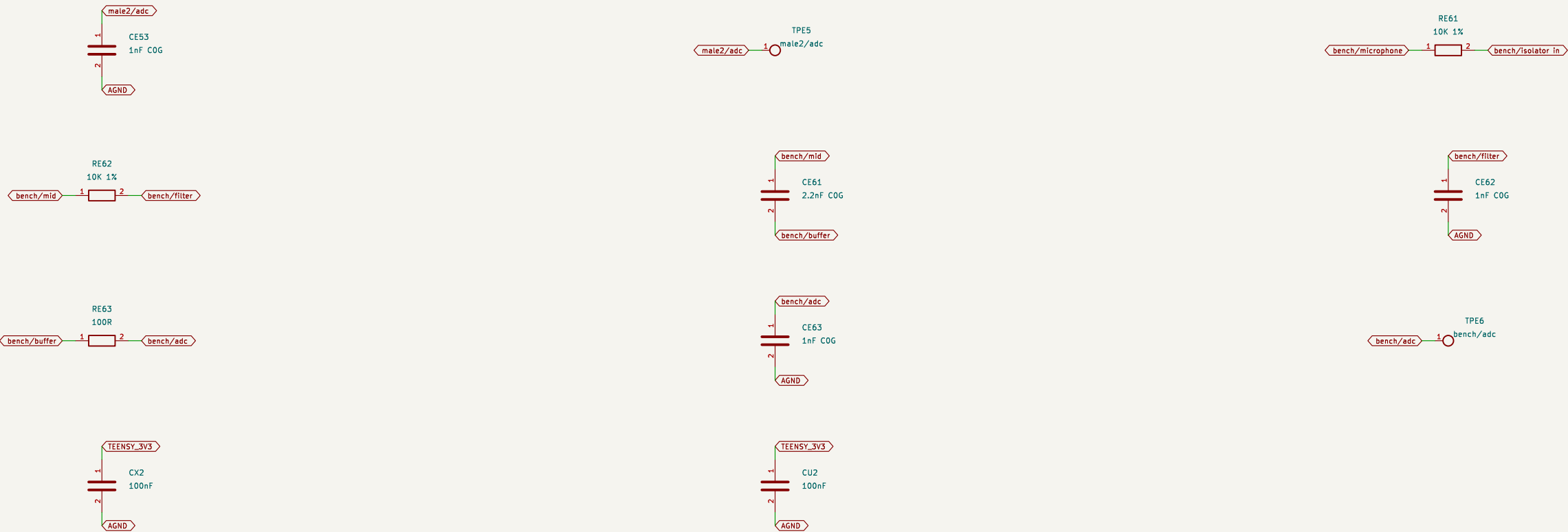
Rev: A-review

KiCad E.D.A. 9.0.7

Id: 30/36

THOMAS / TEENSY – EARS2

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.

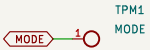
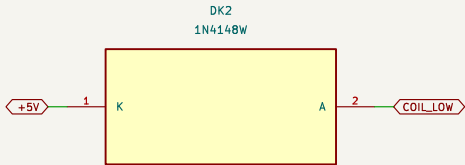
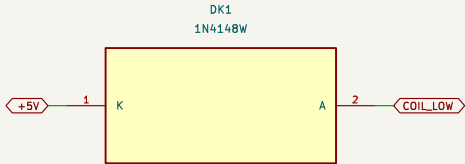
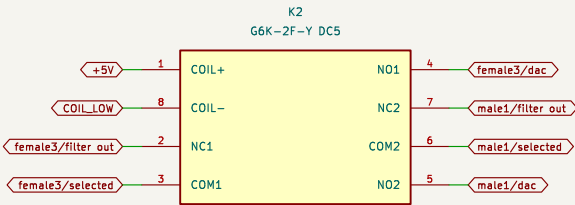
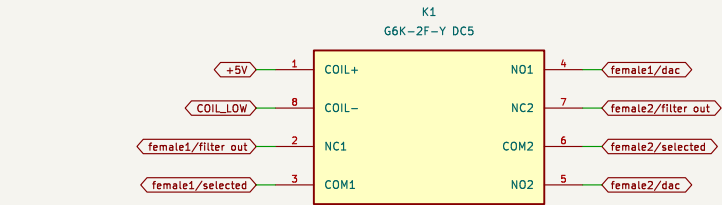


Prototype: verify harness polarity and sensor values before connection
Colloquy of Mobiles
Sheet: /ears2_2/
File: ears2_2.kicad_sch

Title: EARS2		
Size: A3	Date: 2026-10-01	Rev: A-review
KiCad E.D.A. 9.0.7		Id: 31/36

THOMAS / TEENSY – RELAYS

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

Sheet: /relays/
File: relays.kicad_sch

Title: RELAYS

Size: A3
KiCad E.D.A. 9.0.7

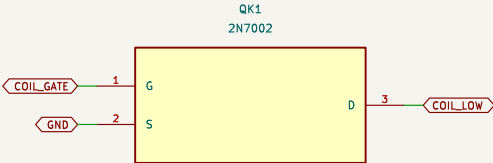
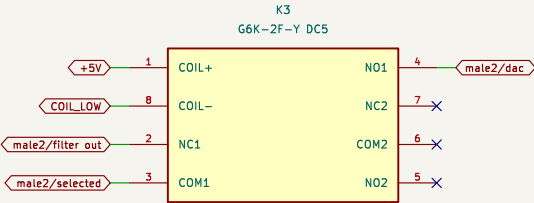
Date: 2026–10–01

Rev: A–review

Id: 32/36

THOMAS / TEENSY – RELAYS

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

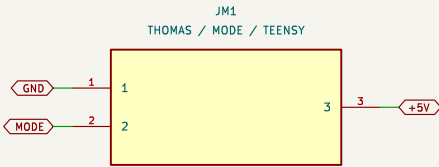
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File: relays_2.kicad_sch

Title: RELAYS

Size: A3	Date: 2026-10-01	Rev: A-review
KiCad E.D.A. 9.0.7	Id: 33/36	

THOMAS / TEENSY – RELAYS

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

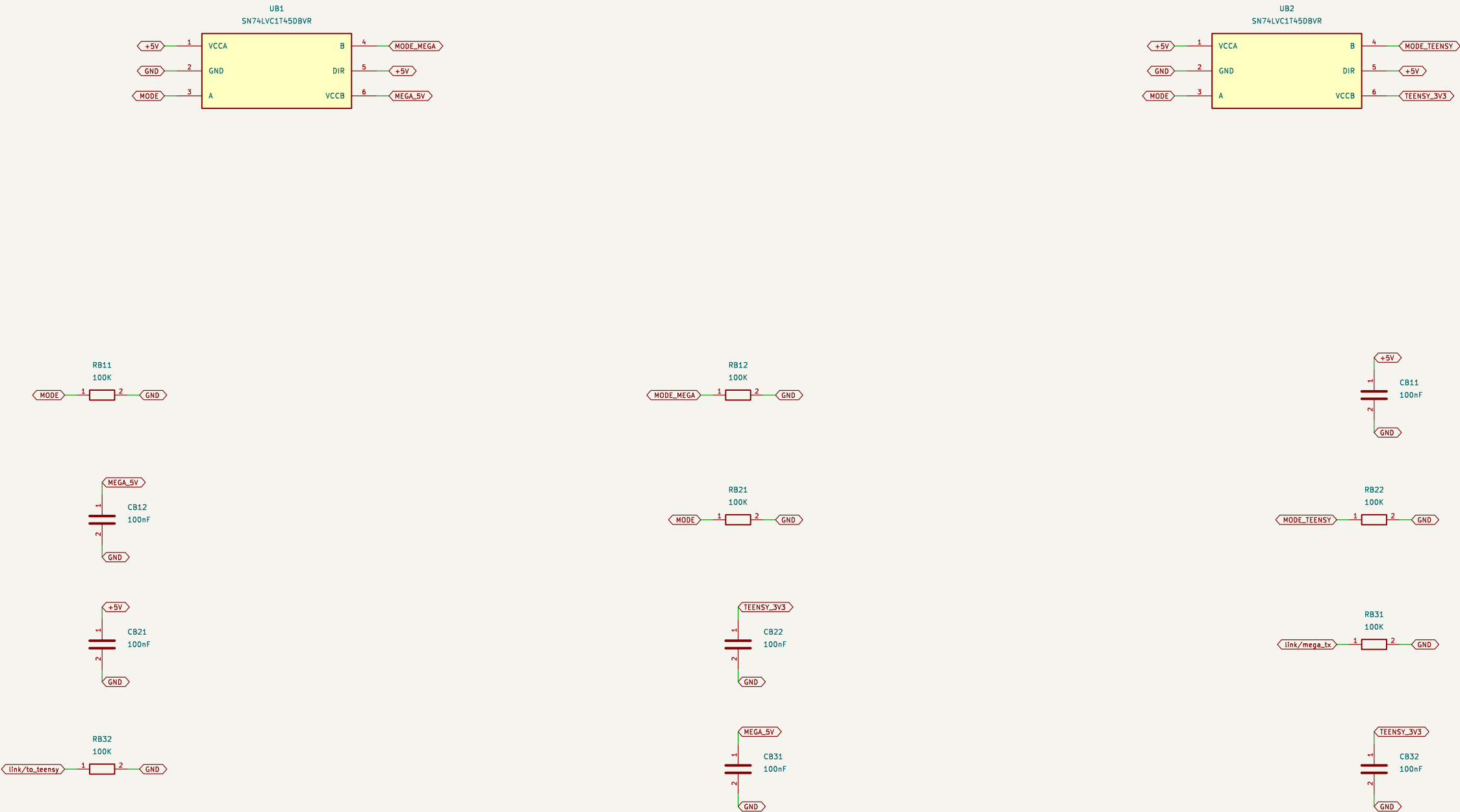
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File: relays_3.kicad_sch

Title: RELAYS

Size: A3	Date: 2026-10-01	Rev: A-review
KiCad E.D.A. 9.0.7	Id: 34/36	

THOMAS / TEENSY – LOGIC

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection

Colloquy of Mobiles

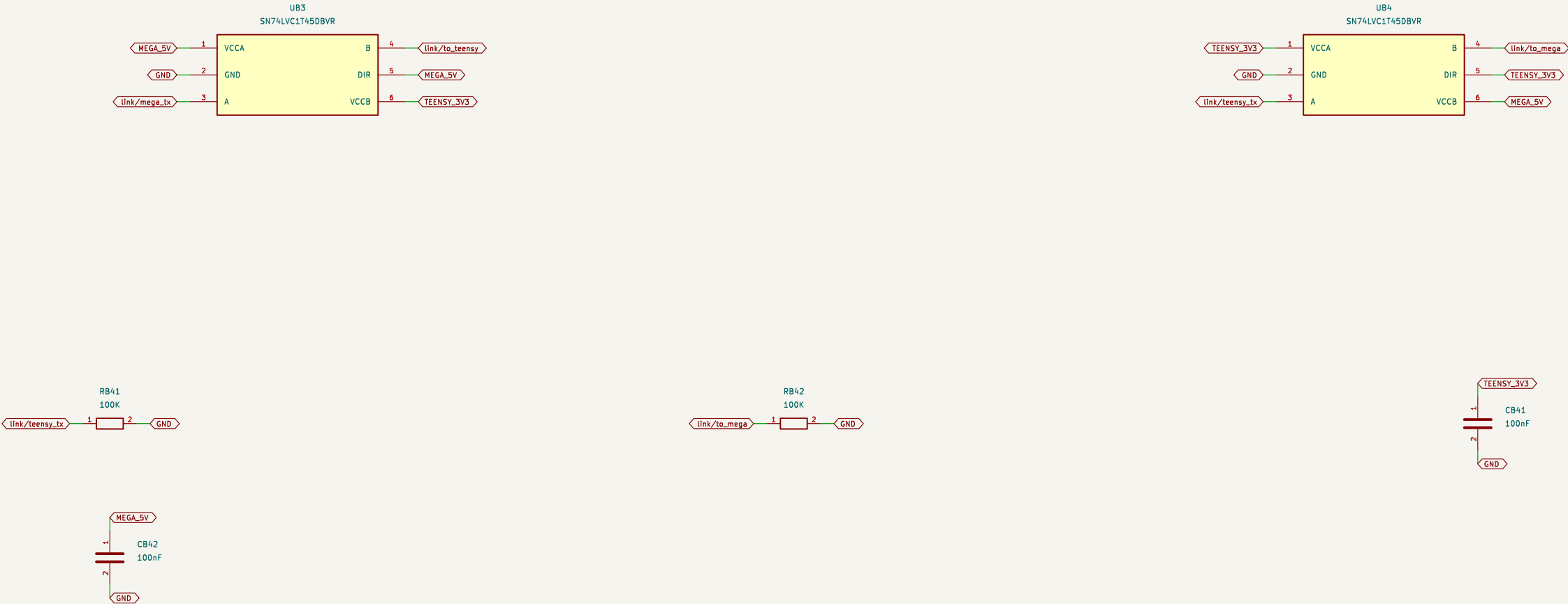
Sheet: /logic/
File: logic.kicad_sch

Title: LOGIC

Size: A3	Date: 2026-10-01	Rev: A-review
KiCad E.D.A. 9.0.7		Id: 35/36

THOMAS / TEENSY – LOGIC

Rev A review | Global net labels join sheets. Pin names and pad numbers are distinct.



Prototype: verify harness polarity and sensor values before connection
Colloquy of Mobiles
Sheet: /logic_2/
File: logic_2.kicad_sch

Title: LOGIC		
Size: A3	Date: 2026-10-01	Rev: A-review
KiCad E.D.A. 9.0.7		Id: 36/36